RIG-SURJECTIVITY OF A CLOSED IMMERSION OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete in both directions

Editorial note. The label lemma-closed-immersion-rig-surjective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- closed immersions of locally Noetherian formal algebraic spaces and their taut surjections B to A in WAdm^Noeth on affine charts
- rig-surjectivity, its base change stability, the monomorphism reformulation and the upshot reduction
- ideals of definition of adic rings and the Jacobson/DVR embedding lemma for one-dimensional quotients

Source reference. Original lemma 00PH: sources/stacks-chow/tags.txt, source lines 931–931. Prerequisite pointer invoked at proof line(s) 5580; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AQI: sources/stacks-chow/tags.txt, source lines 13511–13511. Prerequisite pointer invoked at proof line(s) 5566; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQS: sources/stacks-restricted/pinned-original-source.tex, source lines 5324–5324. Prerequisite pointer invoked at proof line(s) 5569; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AR0: sources/stacks-restricted/pinned-original-source.tex, source lines 5525–5525. Prerequisite pointer invoked at proof line(s) 5584; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQV: sources/stacks-restricted/pinned-original-source.tex, source lines 5394–5394. Prerequisite pointer invoked at proof line(s) 5590; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces. Assume that X and Y are locally Noetherian and f is a closed immersion. The following are equivalent*

(1) *f is rig-surjective, and*
(2) *for every affine formal algebraic space V and every morphism $V \to Y$ which is representable by algebraic spaces and étale the morphism $X \times_Y V \to V$ corresponds to a surjective morphism $B \to A$ in $WAdm^{Noeth}$ whose kernel J has the following property: $IJ^n = 0$ for some ideal of definition I of B and some $n \geq 1$.*

Proof. Let us observe that given V and $V \to Y$ as in (2) without any further assumption on f we see that the morphism $X \times_Y V \to V$ corresponds to a surjective morphism $B \to A$ in $WAdm^{Noeth}$ by Formal Spaces, Lemma 0AQI.

Assume (1). By Lemma 0AQS we see that $\mathrm{Spf}(A) \to \mathrm{Spf}(B)$ is rig-surjective. Let $I \subset B$ be an ideal of definition. Since B is adic, $I^m \subset B$ is an ideal of definition for all $m \geq 1$. If $I^m J^n \neq 0$ for all $n, m \geq 1$, then IJ is not nilpotent, hence $V(IJ) \neq \mathrm{Spec}(B)$. Thus we can find a prime ideal $\mathfrak{p} \subset B$ with $\mathfrak{p} \notin V(I) \cup V(J)$. Observe that $I(B/\mathfrak{p}) \neq B/\mathfrak{p}$ hence we can find a maximal ideal $\mathfrak{p} + I \subset \mathfrak{m} \subset B$. By Algebra, Lemma 00PH we can find a discrete valuation ring R and an injective local ring homomorphism $(B/\mathfrak{p})_{\mathfrak{m}} \to R$. Clearly, the ring map $B \to R$ cannot factor through $A = B/J$. According to Lemma 0AR0 this contradicts the fact that $\mathrm{Spf}(A) \to \mathrm{Spf}(B)$ is rig-surjective. Hence for some n, m we do have $I^n J^m = 0$ which shows that (2) holds.

Assume (2). By Lemma 0AQV it suffices to show that $\mathrm{Spf}(A) \to \mathrm{Spf}(B)$ is rig-surjective. Pick an ideal of definition $I \subset B$ and an integer n such that $IJ^n = 0$. Consider a ring map $B \to R$ where R is a discrete valuation ring and the image of I is nonzero. Since R is a domain, we conclude the image of J in R is zero. Hence $B \to R$ factors through the surjection $B \to A$ and we are done by definition of rig-surjective morphisms. □